Origin of Zircon Megacrysts from Cenozoic Basalts in Northeastern Cambodia: Evidence from U-Pb Age, Hf-O Isotopes, and Inclusions

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ABSTRACT

Zircon megacrysts found in basalts may provide a window into metasomatic processes and element transport in the upper mantle. However, little is known about the formation conditions of these zircons. Here we report U-Pb ages, Hf-O isotope compositions, trace element data, and mineral inclusion analyses for six large fragments of zircon megacrysts from the Gem village in northeastern Cambodia. The zircons have low contents of U, Th, Y, P, and rare earth elements (REEs; 130–288 ppm). The normalized REE pattern is characterized by a steeply rising slope from light REEs to heavy REEs and positive Ce anomalies. Secondary ion mass spectrometer isotopic analyses on the zircon megacrysts yield a mean U-Pb age of 0.98 ± 0.04 Ma (2σ) and homogeneous oxygen isotopic composition with a mean $\delta^{18}O$ value of 5.0‰ ± 0.18‰ (2σ), identical to the $\delta^{18}O$ mantle value. The $^{176}Hf/^{177}Hf$ values range from 0.282970 ± 0.000024 (2σ) to 0.283040 ± 0.000016 (2σ), with ε_{Hf} (0.98 Ma) values of 6.6–9.0, obtained by the laser ablation–multicollector–inductively coupled plasma mass spectrometry technique, further points to the origin of the grains in the depleted mantle. The zircon megacrysts contain apatite inclusions and a reaction rim composed of baddeleyite and Al-Fe-Zr-silicate glass. This leads us to conclude that mantle zircon crystallized during a metasomatic event from phosphate-rich fluids and/or silicate melts enriched in zirconium.

Online enhancements: supplementary tables.

Introduction

Zircon is a mineral of singular importance in earth science (Harley and Kelly 2007). The usefulness of zircon in crustal studies stems from its suitability as a geochronometer based on the decay of U (and Th) to Pb, but zircon is also an important Hf isotopic tracer. Beyond that, zircon oxygen isotopic and rare earth element (REE) and other trace element abundances yield useful clues about the history of the host rock (Corfu et al. 2003). Zircon is a common mineral in continental crustal rocks, especially in igneous rocks of intermediate to acid composition (Hoskin and Schaltegger 2003). However, zircon can also survive under conditions of the earth's mantle. In the past 20 years, there has been a rise of interest in studying mantle-derived zircons that commonly occur as megacrysts in association with alkali basalts and kimberlites (Kinny and Meyer 1994; Valley et al. 1998; Griffin et al. 2000; Zartman and Richardson 2005; Page et al. 2007; Siebel et al. 2009; Sutherland and Meffre 2009; Izokh et al. 2010; Li et al. 2010; Simonetti and Neal 2010; Yu et al. 2010; Abduriyim et al. 2012; Kamenetsky et al. 2014). Similar to crustal zircons, oxygen and hafnium isotope composition, trace element data, and inclusion characteristics have been employed for the analyses of mantle-derived zircons (Belousova et al. 2002; Hoskin and Schaltegger 2003; Thomas et al. 2003; Valley 2003). However, little is known about the formation conditions of such zircons, and the genesis of zircon megacrysts is still controversial (Valley et al. 1998; Page et al. 2007). Moreover, the composition of the

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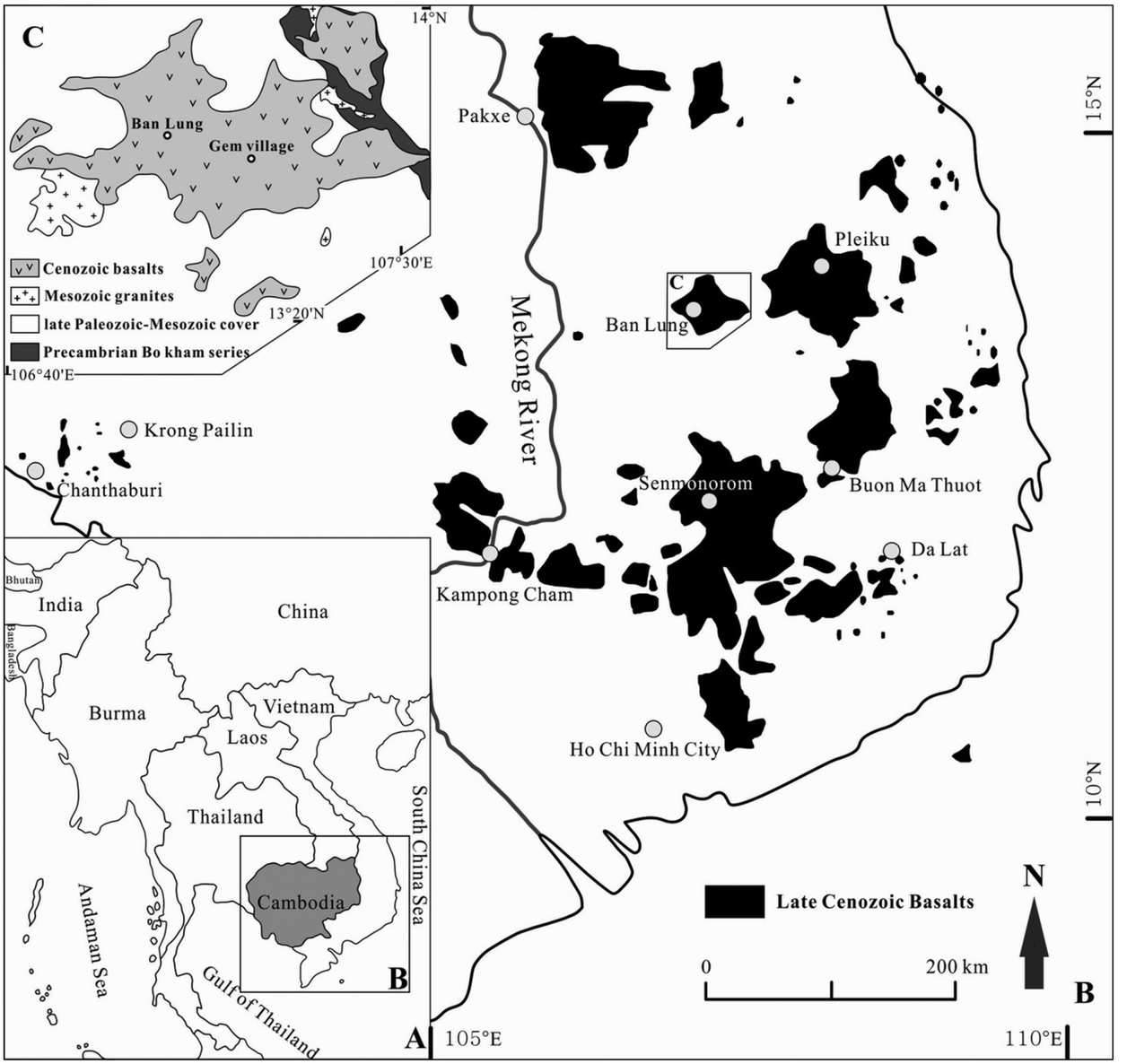

Figure 1. *A*, *B*, Distribution of the late Cenozoic basalts in Southeast Asia (after Barr and Macdonald 1981). *C*, Simplified geological map of study area (after Harada 2010).

megacryst-forming agents and their relationship to the host rocks are still under debate (Griffin et al. 2000; Zartman and Richardson 2005).

Since the late Cenozoic, widespread intraplate volcanism has occurred in East and Southeast Asia. In this region, volcanism has built up a continental basalt province comparable in dimension to the volcanic provinces of eastern Australia or the western United States (Barr and Macdonald 1981; Whitford-Stark 1987). Although numerous studies have been conducted on these rocks (e.g., Barr and Macdonald 1981; Whitford-Stark 1987; Hoang and Flower 1998; Lee et al. 1998), it is still not yet well constrained whether this volcanism is the result of large-scale upwelling and decompression related to mantle plume activity (Tu et al. 1992; Xu et al. 2012) or mantle extrusion (Flower et al. 1998; Niu 2005; Hoang et al. 2013).

Whatever the exact cause of this volcanism, the basalts are a major source of gem-quality corundums, zircons, and sapphire (e.g., Sutherland et al. 2002). Most of these minerals were recovered from placer deposits adjacent to dissected vents. It has been shown that the megacrysts (also including cli-

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nopyroxene, olivine, ilmenite, spinel, magnetite, anorthoclase, plagioclase, nepheline, and garnet) and ultramafic nodules are generally restricted to more alkaline rocks (i.e., nepheline, hawaiite, basanite, and nephelinite; Barr and Dostal 1986). Exploration of the gem-quality corundums, zircons, and other megacrysts is an important economic resource for Southeast Asian countries (Guo et al. 1996; Sutherland et al. 2002; Graham et al. 2008). However, insufficient knowledge about the genesis of the gem-quality minerals and their relationship with the host basalts has greatly limited further exploration strategies for new economic deposits. In this article, we present a comprehensive study—including analysis of in situ U-Pb, Hf, and O isotopes; trace element compositions; and identification of inclusions within zircon megacrysts—on six large (size) fragments of zircon megacrysts from the Gem village in northeastern Cambodia. The chemistry and isotope data of mantle-derived zircons allows us to put new constraints on the origin of zircon megacrysts, and provide further insight into the composition of the upper mantle underneath active volcanic zones.

Geological Setting and Samples

The Southeast Asian basalt plateau was produced by a Neogene-Quaternary intraplate volcanic activity (fig. 1*B*). The generation of this plateau was associated with pull-apart basins and extensional rifts (Barr and MacDonald 1981; Whitford-Stark 1987). Mainland Southeast Asia comprises a giant jigsaw puzzle of continental blocks, volcanic arc terranes, suture zones, and accreted continental crust (Metcalfe 2011). Indochina is the largest tectonic block in Southeast Asia. It is separated from the South China block by the Ailaoshan-Song Ma suture zone and from the Sibumasu block by the Changning-Menglian-Inthanon suture zone (Sone and Metcalfe 2008; Metcalfe 2011). The collision of India with Asia has rotated and extruded Indochina to the southeast along the left-lateral Red River fault zone, with concomitant opening of the South China Sea and Andaman Sea (Tapponnier et al. 1982, 1990). The study area, located in northeastern Cambodia, comprises gneiss and amphibolite of the Precambrian Bo Kham series, late Paleozoic-Mesozoic volcanic and sedimentary rocks, Mesozoic granites, and Cenozoic basalts of the Bokeo Plateau (fig. 1*C*; Harada 2010). The Bokeo basalts show a wide range of compositions ranging from tholeiitic to highly alkalic, including tholeiites, hawaiites, mugearites, alkali olivine basalts, nepheline mugearites, nepheline hawaiites, and basanites (Barr and MacDonald 1981). Early episodes from the middle to late Miocene usually produced quartz and olivine tholeiitic flows, whereas alkaline basalts predominantly emplaced during later episodes from the Pliocene to Quaternary (Hoang and Flower 1998; Lee et al. 1998). Basalt flows cap older volcanic, sedimentary, and metamorphic rocks. Gems—predominantly zircons in association with large crystals of corundum, garnet, titanomagnetite, spinel, and anorthoclase—were found in many of these basalts (Barr and MacDonald 1981; Coenraads et al. 1995).

The zircon megacrysts studied here were sampled with help of the local people on the weathered red-earth surfaces of basalts at the Gem village (fig. 1*C*; lat 13°42.30′N, long 107°11.78′E), northeastern Cambodia. The megacrysts are several to more than 20 mm long and translucent and maroon in color. Minerals with granular, subangular, round, short prismatic, and tetragonal prismatic shapes have been

Figure 2. Morphology of gem zircons. Gem zircons are several to more than 20 mm long, translucent, and maroon in color, with granular, subangular, round, short prismatic, and tetragonal prismatic shapes.

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found (fig. 2). Zircon crystals show no magmatic oscillatory zoning in cathodoluminescence (CL) images (fig. 3) and are homogeneous in backscattered electron (BSE) images (fig. 4*A*). Six representative zircon megacrysts were analyzed in this study.

Analytical Methods

Zircon U-Pb Isotopes. Zircon shards together with the Penglai reference zircon were placed in epoxy resin mounts and then polished. All zircons were documented with transmitted and reflected light micrographs as well as CL and BSE images to reveal their internal structures, and the mounts were vacuum coated with high-purity gold. Measurements of U, Th, and Pb isotopes were conducted by the secondary ion mass spectrometer (SIMS) technique using a Cameca IMS-1280 ion microprobe at the Institute of Geology and Geophysics, Chinese Academy of Sciences (Beijing). The analytical procedures are described by Li et al. (2010). The Penglai zircon was alternatively analyzed as reference material and yielded a mean age of 4.3 ± 0.1 Ma (MSWD = 1.8, n = 14, 2σ), which is identical within errors with the recommended value of 4.4 ± 0.1 Ma (Li et al. 2010). Because the analyzed zircons are very young (ca. 0.98 Ma) and have low contents of radiogenic Pb, both ^{207}Pb-corrected ^{206}Pb/^{238}U ages and ^{230}Th-corrected ^{206}Pb/^{238}U ages are given in table S1 (tables S1–S4 available online). Uncertainties on individual analyses are reported at the 1σ level. Mean ages for pooled ^{206}Pb/^{238}U results are quoted with 95% confidence intervals.

Zircon Oxygen Isotopes. Zircon oxygen isotopes were measured by the SIMS technique at the Institute of Geology and Geophysics, Chinese Academy of Sciences (Beijing), with analytical procedures similar to those reported by Li et al. (2010). After U-Pb dating, the sample mount was reground and repolished to ensure that any oxygen implanted in the zircon surface from the O_2 beam used for the U-Pb determination was removed. The SIMS oxygen isotopic determinations were done on the same zircon shards. ^{18}O/^{16}O ratios are expressed as δ^{18}O and reported versus the Vienna

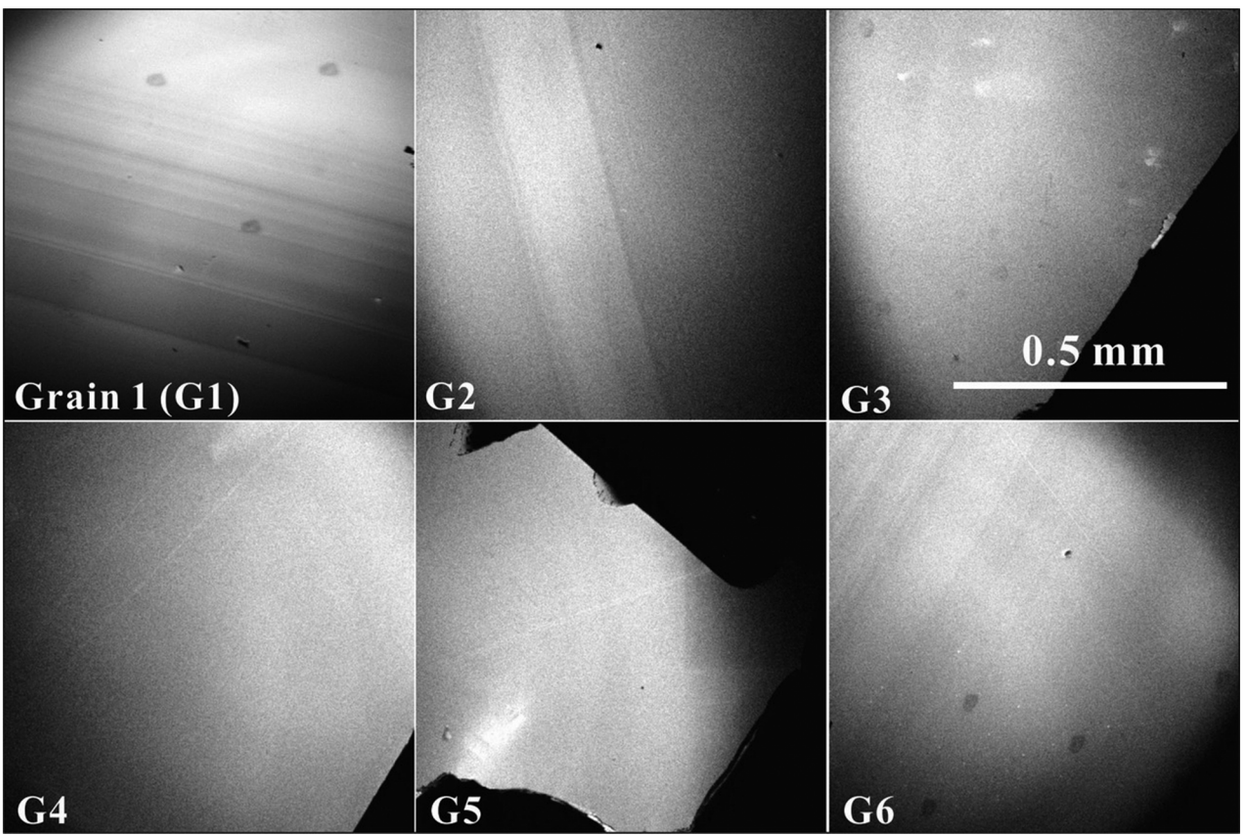

Figure 3. Cathodoluminescence (CL) images of zircon megacrysts. Zircons show poorly developed growth zoning on CL images. A color version of this figure is available online.

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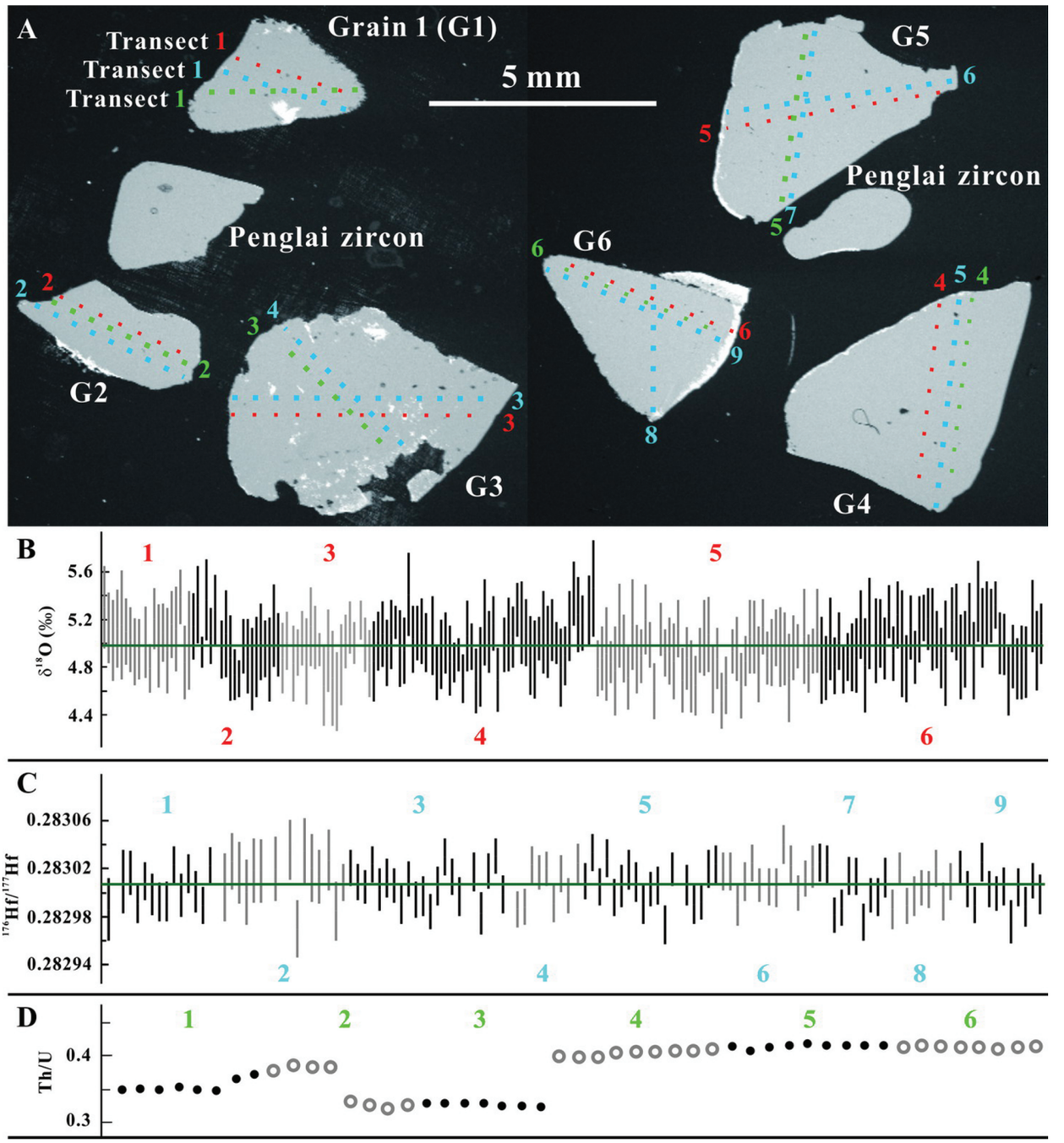

Figure 4. Backscattered electron images (*A*) and profile analyses of oxygen (*B*), hafnium isotope (*C*), and trace element (*D*) measurements and transecting different domains of zircon megacrysts. The red dashed line is the transect of O isotope measurements. The blue and green dashed lines are the transect of Hf isotope and trace element measurements, respectively.

standard mean ocean water (VSMOW) standard: $[(^{18}O/^{16}O)_{sample}/(^{18}O/^{16}O)_{VSMOW} - 1] \times 10^3$. The Penglai zircons yield a weighted mean for $\delta^{18}O$ of 5.25‰ ± 0.04‰ (MSWD = 1.4, $n = 64$, 2σ), which is close to the reported value of 5.31‰ ± 0.1‰ (2σ; Li et al. 2010).

Zircon Hf Isotopes. Hafnium isotopic determination was performed on a Neptune multicollector ICP-MS equipped with a Geolas-193 laser ablation system (laser ablation–multicollector–inductively coupled plasma mass spectrometry [LA-MC-ICPMS]) at the Institute of Geology and Geophysics, Chinese Acad-

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emy of Sciences (Beijing). The analytical procedures are similar to that described by Wu et al. (2006). A total of 128 measurements were made randomly on the six zircon shards that had been previously analyzed for U-Pb and O isotopes. Measured $^{176}Hf/^{177}Hf$ ratios were normalized to $^{179}Hf/^{177}Hf = 0.7325$. Measured $^{176}Hf/^{177}Hf$ ratios for the Penglai zircon (0.282927 ± 0.000006, MSWD = 4.9, $n = 42$, 2σ) are close to the reported value (0.282906 ± 0.000001, 2σ; Li et al. 2010).

Zircon Trace Element Contents. Trace elements of zircons were analyzed on the GeoLasPro laser ablation system and Agilent7500a inductively coupled plasma mass spectrometer (LA-ICPMS) at the Key Laboratory of Crust-Mantle Materials and Environments at the University of Science and Technology of China (USTC), Hefei. The analyses were carried out with a spot diameter of 160 μm and an ablation depth of ~20–40 μm, with He as the carrier gas and a 6-Hz repetition rate. We used zircon 91500 as an external reference material. Element concentrations of zircon were calibrated against the NIST610 reference material, which was measured twice every 10 sample spots, using Si as an internal standard. The trace elemental raw data were processed with the Glitter 4.0 (Macquarie University, Australia) program.

Inclusion Studies. After trace element and isotope analysis, the sample surface was repolished for laser Raman identification at the Key Laboratory of Crust-Mantle Materials and Environments at USTC (Hefei) and scanning electron microscopy (SEM) analyses at Chengdu Center of China Geological Survey (Chengdu). Inclusions in zircons were observed under an optical microscope and identified by laser Raman spectroscopy (ThermoFisher DXR) with a 532-nm line of the laser. A ×50 microscope objective was used to focus the laser onto the sample and collect the Raman signal. The SEM analyses were conducted using a Hitachi S-4800 apparatus, with an accelerating voltage of 15 kV. The system was equipped with secondary and BSE detectors and a silicon drift energy dispersive spectrometer (EDS).

Results

U-Pb Age. U-Pb isotope measurements of 49 spots on six zircon megacrysts are given in table S1. Th concentrations are low and range from 4 to 23 ppm and U from 13 to 55 ppm. Th/U ratios are mostly between 0.3 and 0.4 (table S1). The zircons give a ^{230}Th-corrected $^{206}Pb/^{238}U$ weighted mean age of 0.98 ± 0.04 Ma (fig. 5; MSWD = 1.05, 2σ). No differences are noted between core and rim results, suggesting uniform ages across the crystals and generally uninterrupted growth. The obtained U-Pb zircon age lies within the eruptive age spectrum for late Cenozoic basaltic volcanism in Southeast Asia (Barr and Macdonald 1981; Lee et al. 1998).

Oxygen Isotope Data. SIMS measurements of oxygen isotopes were conducted on the same six zircon grains (fig. 4*A*), and the data are given in table S2. Values of $\delta^{18}O$ range from 4.5‰ to 5.5‰, with a weighted mean value of 5.0‰ ± 0.18‰ (2σ; fig. 4*B*). Overall, both the intergrain individual and intragrain profile measurements indicate that zircon megacrysts have fairly homogeneous oxygen isotope values, and the composition is consistent with the value of 5.3‰ ± 0.3‰ (2σ) reported for mantle zircons (Valley et al. 1998).

Hf Isotope Data. Hf isotopic composition of zircon was carried out on the same grains (fig. 4*A*) used for U-Pb and oxygen analyses. The analytical data are given in table S3. No significant differences occur across the crystals, and the zircon megacrysts are found to be fairly homogeneous in Hf isotopes. The measured $^{176}Hf/^{177}Hf$ values range from 0.282970 ± 0.000024 (2σ) to 0.283040 ± 0.000016 (2σ), with a weighted mean of 0.283007 ± 0.000003 (fig. 4*C*; MSWD = 3.1, $n = 121$, 2σ). The $\varepsilon_{Hf}(t)$ values of zircon megacrysts, when calculated back to 0.98 Ma, range from 6.6 to 9.0 (fig. 6). These values are comparable to those of zircon megacrysts in eastern China (Yu et al. 2010) and eastern Australia (Abduriyim et al. 2012).

Trace Element Composition. Results are tabulated in table S4. Chondrite-normalized REE and trace element patterns of zircons are shown in figure 7. All analyzed spots of the six zircon megacrysts show consistent variations in profiles across the crystals.

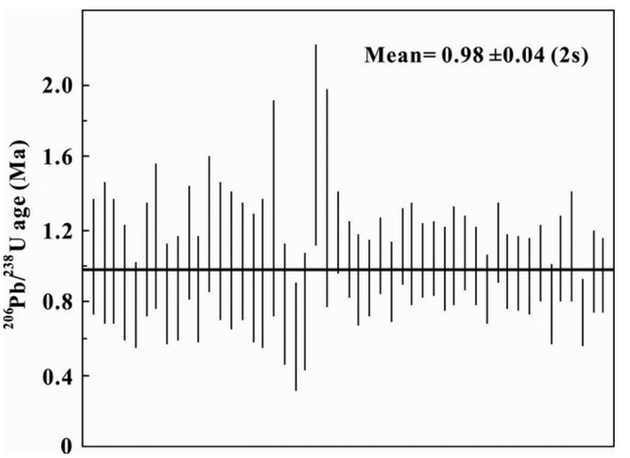

Figure 5. U-Pb isotopic analyses of zircon megacrysts. ^{230}Th-corrected $^{206}Pb/^{238}U$ weighted mean age is 0.98 ± 0.04 Ma (95% confidence interval, MSWD = 1.05, 2σ).

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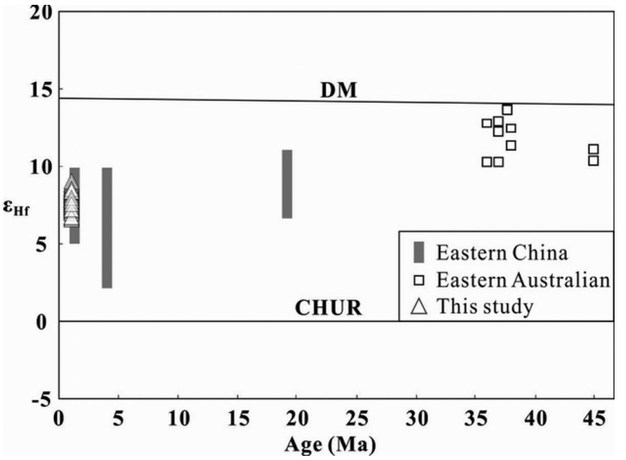

Figure 6. ε_{Hf} of the zircon megacrysts from Cambodia, eastern China (after Yu et al. 2010), and eastern Australian (after Abduriyim et al. 2012), plotted against the crystallization age of the zircons. Parameters of depleted mantle (DM) are from Nebel et al. (2014).

SiO_2 ranges from 28.7 to 30.0 wt% and ZrO_2 amounts at 66.1 wt%. HfO_2 concentrations range from 0.71 to 0.79 wt%, with a mean of 0.74 wt%. Yttrium concentrations fall in the range from 197 to 437 ppm, and phosphorus ranges from 50.5 to 79.3 ppm. The total REE (∑REE) concentrations are relative low, that is, between 130 and 288 ppm. The chondrite-normalized REE patterns show variable enrichment of heavy REEs, positive cerium, and no europium anomalies in all analyzed grains (fig. 7*A*). These features are similar to zircon megacrysts derived from alkali basalts but different from zircons from kimberlitic or granitic intrusions (fig. 7*A*). The mean abundances of many trace elements and the shape and slope of normalized trace element patterns are distinctive for zircons from different sources (fig. 7*B*). There is a general trend of increasing trace element abundance in zircons from kimberlite through basalt and carbonatite to granitoid rocks (fig. 7*B*).

Inclusions. One zircon grain studied here contains abundant mineral inclusions (fig. 8). We have applied laser Raman spectroscopy to identify their compositions. Our results show that the mineral inclusions mainly consist of apatite (figs. 8, 9). The apatite inclusions are granular, droplet, ovoid to skeleton shaped, and usually nucleated and accumulated within the tubular channels and bifurcating cavities (fig. 8). The tubular channels (10–80 μm wide) are found to be empty and orientated parallel to each other (fig. 8*A*). In some small apatite inclusions situated outside the tubular channels, vapor bubbles were identified (fig. 8*A*).

Representative Raman spectra of phases within the studied inclusions are shown in figure 9. Raman peaks at 202, 214, 225, 356, 439, 974, and 1008 cm^{-1} correspond to host zircon. The peaks in the spectra at 450, 560, 583, 632, and 961 cm^{-1} are described as phosphate vibrational modes of apatite by Penel et al. (1998), Antonakos et al. (2007), and Ulian et al. (2013). The v_2 PO_4 mode shows one peak at 450 cm^{-1}; the v_4 PO_4 mode shows peaks at 560, 583, and 632 cm^{-1}; and the v_1 PO_4 mode shows one peak at 961 cm^{-1} (fig. 9), while two unidentified Raman peaks appear at 506 and 530 cm^{-1}.

We also found a 500-μm-wide multiple-phase reaction rim with an irregular shape within another zircon grain. BSE image and corresponding EDS spectra of this reaction rim are shown in figure 10. The reaction rim consists of baddeleyite and Al-Fe-Zr-silicate glass. Baddeleyite with an irregular granular, skeletal, and needle shape mainly occurs along the boundary of the reaction rim. The worm- and oolitic-shaped silicate glass and baddeleyite are intergrown with each other. Some resin was also found

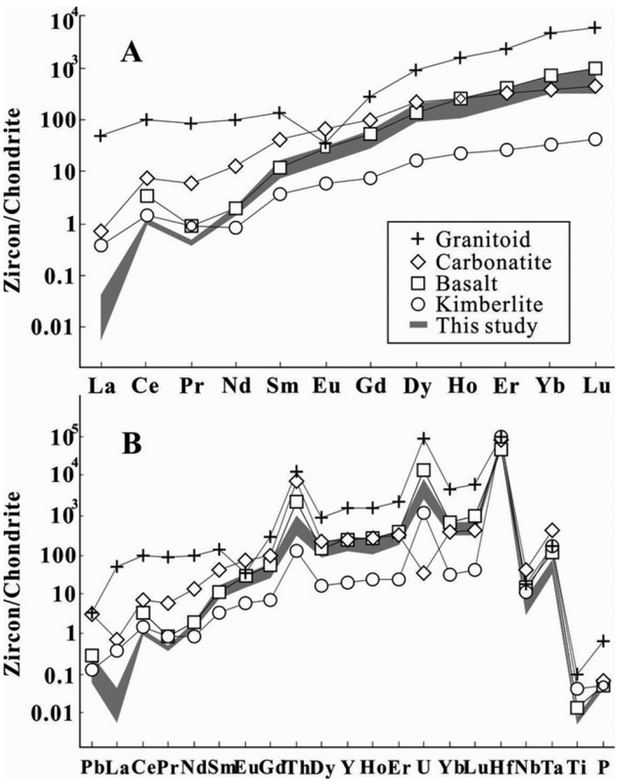

Figure 7. Chondrite-normalized rare earth element (*A*) and trace element (*B*) patterns of zircons from different rock types. Data of this study are from table S4, available online. Data of granitoid, carbonatite, basalt, and kimberlite are from Belousova et al. (2002). Chondrite values are from Sun and McDonough (1989).

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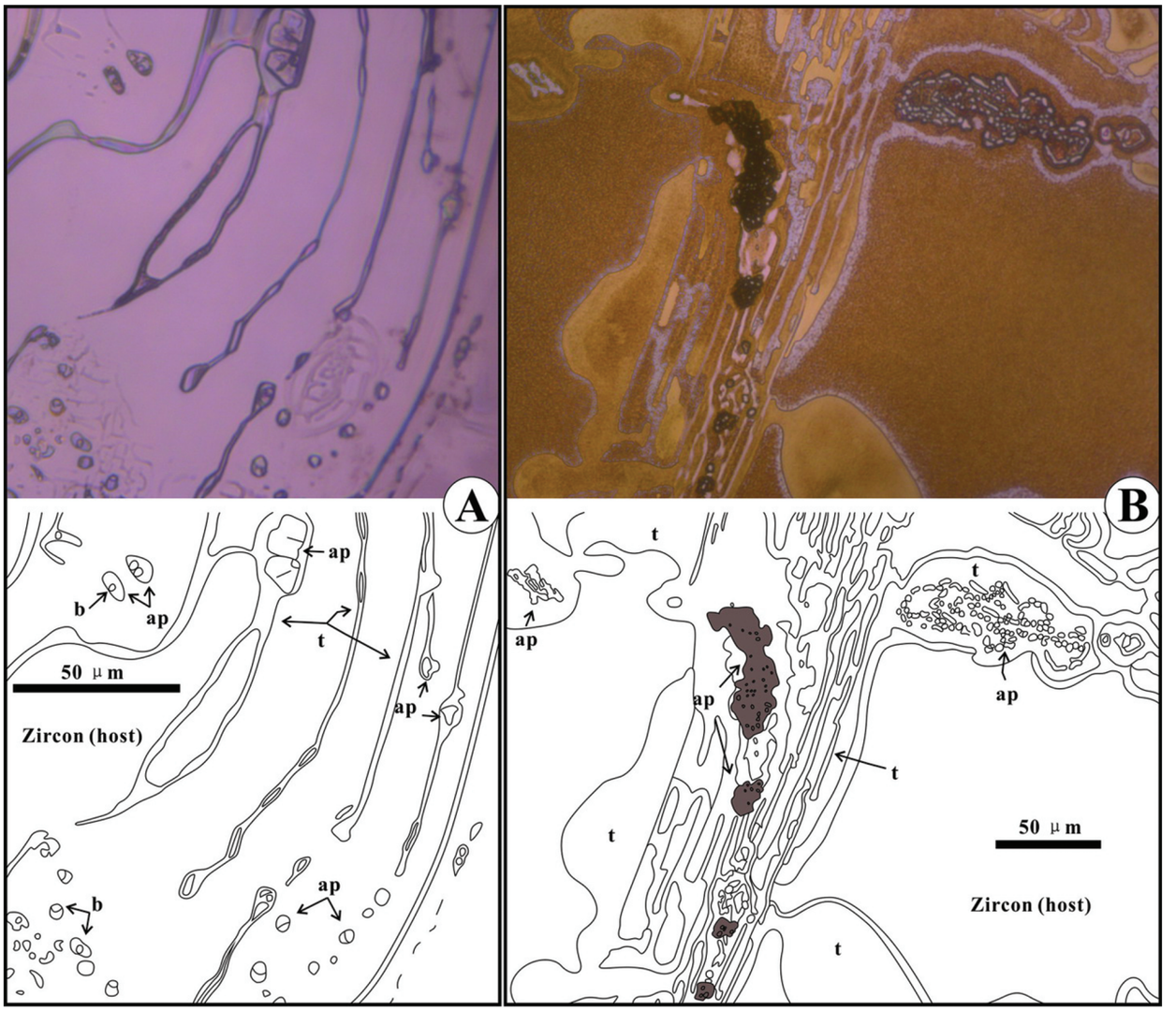

Figure 8. Transmitted light photographs of mineral inclusions in zircons. *A*, Inset shows granular apatite (ap) inclusions, which occur beside and within the tubular channels (t). The tubular channels (10–80 μm wide) were found to be empty and parallel to each other with the same orientation. Some granular apatites contain vapor bubbles (b). *B*, Inset shows granular, ovoid, and skeleton-shaped apatite inclusions within the tubular channels and bifurcating cavities.

in the reaction rim, which was probably introduced when the zircons were placed in epoxy resin mounts.

Discussion

Mantle Affinity of Zircon Megacrysts. Usually mantle-related zircons are either coeval with their host basalts and kimberlites (Visonà et al. 2007; Graham et al. 2008; Yu et al. 2010; Abduriyim et al. 2012; Kamenetsky et al. 2014) or crystallized and resided in the mantle long time before their entrainment into magma and eruption to the surface (Konzett et al. 1998; Schmidberger et al. 2005; Zartman and Richardson 2005; Siebel et al. 2009). Although exposed to mantle temperatures, inheritance of U/Pb isotopic ratios is well documented in mantle zircons, because of their low U and Th contents, which limit the degree of radiation damage to the lattice and hence the rate of Pb diffusion (Zartman and Richardson 2005). The CL and BSE images of those zircon megacrysts show homogeneous characteristics, and the Hf-O isotopic as well as the trace elemental compositions are almost consistent (figs. 4, 7), which further imply that metamic-

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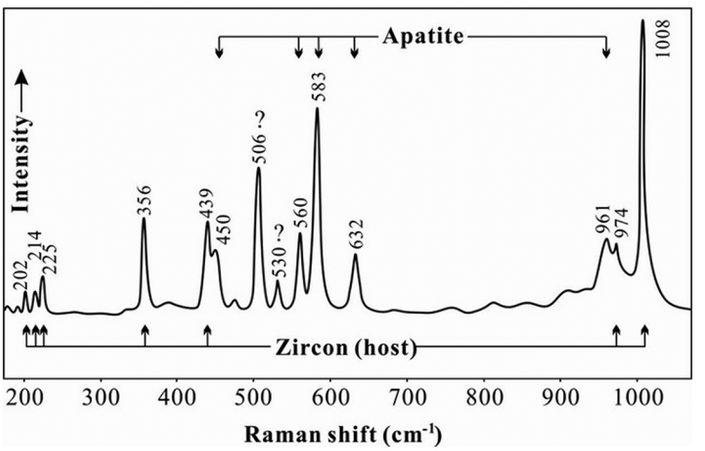

Figure 9. Representative Raman spectra of phases within the studied inclusions. All spectra were obtained at room temperature. Raman spectrum showing peaks at 202, 214, 225, 356, 439, 974, and 1008 cm^{-1} corresponding to host zircon and peaks at 450, 560, 583, 632, and 961 cm^{-1} corresponding to apatite; peaks at 506 and 530 cm^{-1} are unidentified.

tization would be not important. So compositions of the zircon megacrysts could reflect characteristics of parental magma, and we suggest that the U-Pb ages (ca. 0.98 Ma) reported here represent the crystallization age of the zircons, which is consistent with the eruption time from Pliocene to Quaternary of alkaline basalts in northeastern Cambodia and south-central Vietnam (Hoang and Flower 1998; Lee et al. 1998).

The zircon megacrysts from northeastern Cambodia investigated in this study are generally large (millimeter to centimeter size), irregular or rounded in shape, clear, and gemmy, and some of them show abrasion on the surface. Thus, the grains are clearly texturally distinguishable from crustal zircons. Zircons from the Earth mantle usually show relative low U, Pb, Th, P, Y, and REE abundances (Heaman et al. 1990; Belousova et al. 2002; Hoskin and Schaltegger 2003), and zircon crystals investigated here share these features (table S4), whereas zircons from crustal rocks are more enriched in these elements (Pupin 2000; Belousova et al. 2002). A number of studies have shown that zircons of mantle and crustal affinity also have distinct trace elements and REE compositions (Belousova et al. 2002; Hoskin and Schaltegger 2003). Zircon megacrysts studied here show chondrite-normalized REE patterns (fig. 7*A*), with a steep slope from light REEs (LREEs) to heavy REEs, positive Ce anomalies, and absence of Eu anomalies, similar to mantle-derived zircons (Belousova et al. 2002; Hoskin and Schaltegger 2003; Siebel et al. 2009). When Ce^{3+} is oxidized to Ce^{4+}, it behaves very similar to Zr or Hf, and thus it is preferentially incorporated into the zircon lattice over the other LREEs (Hinton and Upton 1991). The Eu anomaly may be explained by plagioclase fractionation, which depletes Eu from the magma before or during zircon crystallization (Snyder et al. 1993; Hoskin and Schaltegger 2003). Hence, the relatively low REE concentration, lack of Eu anomaly, and other trace element concentrations are compatible with the view that zircons crystallized from a plagioclase-free mantle source.

Positive $\varepsilon_{Hf}(t)$ values of 6.6 to 9.0 of these zircon megacrysts also suggest a mantle affinity. Moreover, the $\delta^{18}O$ mean value of 5.0‰ ± 0.18‰ is identical to the normal mantle value of 5.3‰ ± 0.3‰ (Valley et al. 1998). Such isotopic characteristics have been reported for other zircons argued to have been formed in the mantle (Griffin et al. 2000; Qiu et al. 2005; Siebel et al. 2009; Yu et al. 2010; Abduriyim et al. 2012). On the basis of this evidence, zircon megacrysts from Cenozoic basalts in northeastern Cambodia should have been derived from a mantle source. The poorly developed growth zoning (CL images; fig. 3) may be related to the residence of the megacrysts at high temperatures in the mantle, which may have led to partial or complete homogenization of any original primary zoning (Corfu et al. 2003). Considering the fact that the range of 2.4 ε_{Hf} units in Hf isotopes is a bit large to be a reflection of the analytical precision, we estimate that the 2.4 ε_{Hf} units in Hf isotopes are related to a blend of depleted mantle with metasomatic agent (see below).

Nature of the Megacryst-Forming Magma. The mantle origin of the zircon megacrysts is apparent

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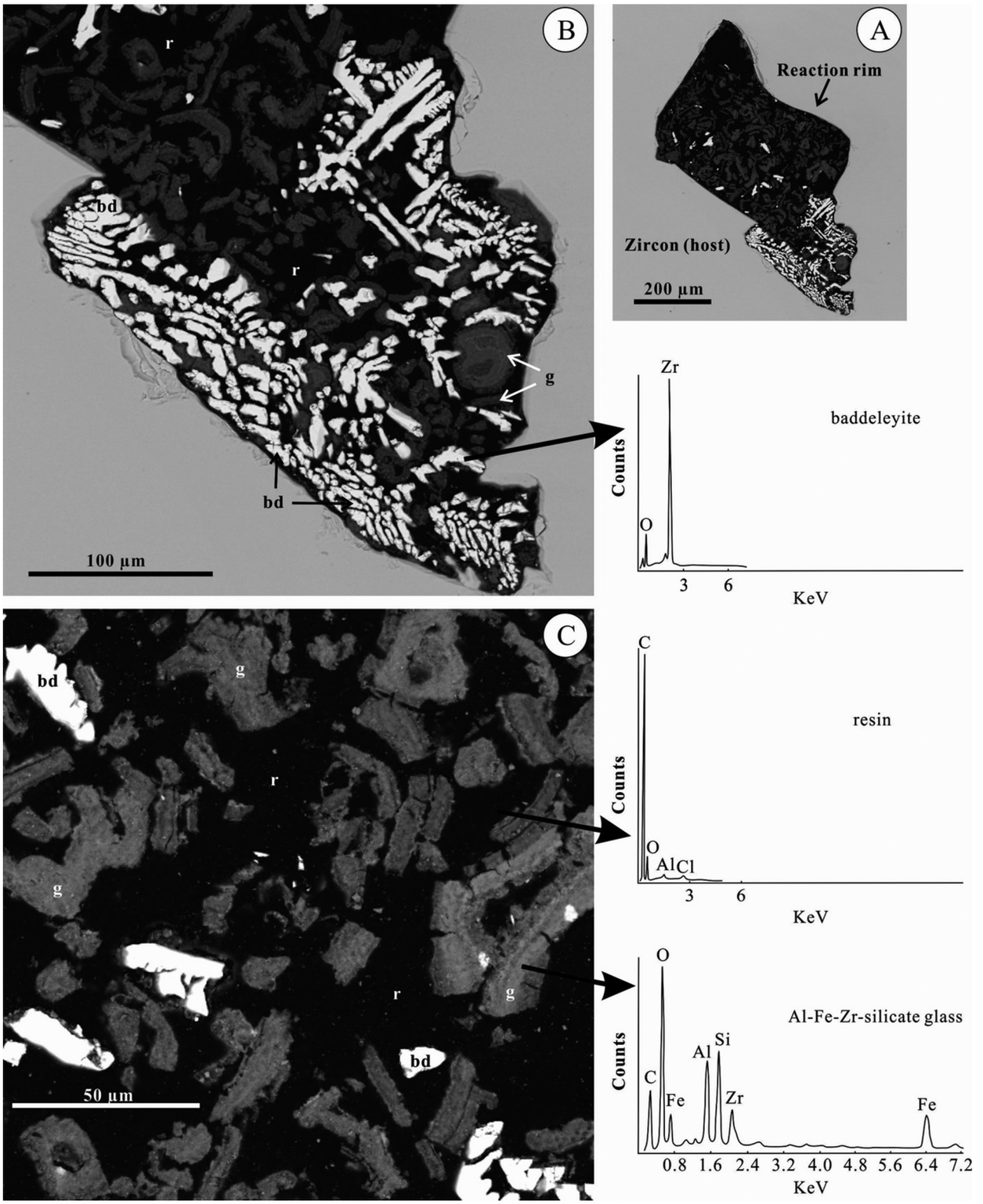

Figure 10. Backscattered electron image and corresponding energy dispersive spectrometer spectra of the reaction rim in host zircon. *A*, Inset shows a 500-μm-wide reaction rim with an irregular shape within zircon. The reaction rim consists of baddeleyite (white; bd), Al-Fe-Zr-silicate glass (gray; g), and resins (black; r). Details shown in *B* and *C*. *B*, Inset shows irregular granular, skeletal, and needle-like baddeleyites that mainly occur along the boundary of the reaction rim. *C*, Worm- and oolitic-shaped silicate glass and baddeleyites are intergrown with each other. A color version of this figure is available online.

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from their morphology, trace element compositions, and Hf-O isotope composition. However, the nature of the megacryst-forming agent is still unclear. The fact that U-Pb ages for these megacrysts mostly correspond to the eruption time of the host magma suggests a genetic link between zircon and host basalt, despite the common observation that zircon megacrysts show reaction rims of ZrO_2 (Heaman and LeCheminant 1993; Corfu et al. 2003; Kamenetsky et al. 2014), which implies that they were not in equilibrium with the mafic-ultramafic host magma (Kinny and Maas 2003). Although there are no available Nd isotope data of the zircons, their Hf isotopes (fig. 6) are slightly more depleted compared with the basalts in northeastern Cambodia and south-central Vietnam, which have ε_{Nd} values of 1.5–4.3 (Hoang and Flower 1998), indicating that the zircons did not directly crystallize from the basaltic magma. The zircon megacrysts are commonly associated with the alkaline basaltic rocks in this area but not tholeiitic flows. The former originated from the asthenospheric mantle and the latter likely from the lithospheric mantle (Hoang and Flower 1998). It is estimated that zircon megacrysts formed at the base of the lithospheric mantle, so it could be transported by the asthenosphere-sourced magma to the earth surface. Griffin et al. (2000) and Spetsius et al. (2002) argued that zircon megacrysts in kimberlites crystallized from a highly evolved liquid and represent the late-stage crystallization products of an OIB-type magma source. However, the exact evolution processes of a primitive alkaline magma toward a zirconium rich silicate melt from which zircon can precipitate are still not fully understood.

The inclusions in the zircons can provide further clues as to the nature of the mantle source and the conditions of crystallization. We found numerous apatite inclusions within the zircon megacrysts. Apatite is a major site for REE and some LILE and is common as a mantle component of metasomatic origin (O'Reilly and Griffin 2000). Our study confirms the common presence of apatite-rich fluids. Interestingly, apatite inclusions show manifold morphologies—including granular, droplet, and ovoid to skeleton-shaped forms—and are usually accumulated within tubular channels, bifurcating cavities, and vesicles (fig. 8). This phenomenon is explained by the fact that the apatite-rich fluids were trapped during zircon crystal growth and then quickly transported to the earth surface before they burst to form crystals and tubular channels during the decompression and cooling processes. Tubular channels, bifurcating cavities, and vesicles were formed by the exsolution of volatiles from a fluid phase, mainly during rapid quenching. The glassy apatites nucleated and accumulated from the fluid during cooling (fig. 8), and their different morphologies probably reflect differences in the cooling rate, size, and composition of the trapped fluids.

We also found reaction rims within the zircons consisting of baddeleyite and Al-Fe-Zr-silicate glass. The worm- and oolitic-shaped silicate glass and baddeleyites are intergrown with each other (fig. 10). As pointed out by Heaman and LeCheminant (1993) and Corfu et al. (2003), reaction between zircon and other silicates can lead to the formation of baddeleyite. These reactions have been linked to metasomatic processes in the mantle (Heaman and LeCheminant 1993) or late-stage contact between zircon megacrysts and ascending magma (Corfu et al. 2003). The commonly observed textures of reaction rim in mantle zircon have also been reported by Page et al. (2007) and Kamenetsky et al. (2014). From petrographic features, we consider that baddeleyite formed by the reaction between zircon and Al-Fe-Zr-silicate undersaturated melt. The process can be described as follows: $ZrSiO_4$ + Al-Fe-Zr-silicate undersaturated = ZrO_2 + Al-Fe-silicate.

Subcontinental lithospheric mantle has undergone metasomatic modification since its stabilization (O'Reilly and Griffin 2006). Silicate to carbonate magmas in off-craton regions contain varying types and abundances of dissolved metasomatic fluids and solutes, including brines, C-O-H species, and sulfur-bearing components (O'Reilly and Griffin 2013). Although mantle metasomatic processes are far from being understood, the presence of apatite inclusions in this study indicates that phosphate-rich fluids may have been responsible for an environment favoring zircon formation. In addition, reaction rims composed of baddeleyite and Al-Fe-Zr-silicate glass further confirm mantle metasomatic processes related to zirconium rich silicate melts. Griffin et al. (1996) studied garnet porphyroblasts from peridotite in southern Africa, which commonly have rims that are strongly enriched in Fe, Ti, Y, and Zr relative to the cores. They considered that the garnet rims represent metasomatic addition of garnet from a more depleted protolith. These findings support our evidence for the presence of Zr-rich silicate melts in the mantle. Mantle melts/fluids can move and interact with the mantle wall rock through grain-boundary infiltration and crack propagation and successively change in composition as a result of exchange reactions with the mantle wall rock and mineral precipitation processes (O'Reilly and Griffin 2013). Much of the REEs and some large ion lithophile elements in peridotitic wall rocks and zircon-forming magma could be supplied by phosphate-rich fluids through reactions

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among them. Mantle melts/fluids moving through the lithosphere could interact with mantle peridotite and evolve toward zircon forming compositions. Such processes may have taken place in response to the Cenozoic basaltic volcanism in Southeast Asia.

Conclusions

SIMS U-Pb ages of the zircon megacrysts from northeastern Cambodia are consistent with the eruption time of late Cenozoic basaltic volcanic rocks in Southeast Asia. Typical geochemical characteristics—such as low U, Th, Y, and REEs and the absence of negative Eu anomalies, together with the depleted hafnium and mantle-like oxygen isotopic compositions—suggest that the zircon megacrysts formed in the mantle. Petrographic observations and zircon inclusion analyses provide direct evidence for the presence of phosphate-rich fluids and Zr-enriched silicate melts in the mantle, which may have been responsible for an environment favoring zircon crystallization.

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